



SDEV2501

Day 05 - Open Data

A LEADING POLYTECHNIC COMMITTED TO YOUR SUCCESS

Today's objective

- Ingest and preview a simple dataset using open-source tools

REVIEW

**WE
ARE** **ESSENTIAL
TO ALBERTA**



Define

- Range
- IQR
- Variance
- Standard deviation
- Quartiles & Percentile
- Histogram
- Outlier

DATA PORTALS

**WE
ARE** ESSENTIAL
TO ALBERTA



Where do data come from?



Sometimes data are collected **internally**, or accessible through an **API**.



Sometimes: those data have **already** been collected in an external, structured way.

Data Portals

A data portal is:

- A **centralized** dataset catalogue
- **Searchable** and **documented**
- Intended for **reuse**

A data portal is **not**:

- Raw APIs
- Live dashboards
- Perfectly clean data

How do Data Portals support data science?

Data portals support:

- **Evidence-based decisions**
 - e.g. usage patterns
- **Benchmarking**
 - e.g. market trends
- **Context and validation**
 - e.g. demographics, policy impacts

What kind of data can we find?

Demographics

Transportation

Housing

Environment

Economics

Data Portal examples

- data.calgary.ca
- data.edmonton.ca
- open.toronto.ca
- opendata.vancouver.ca/pages/home
- open.alberta.ca/opendata
- data.novascotia.ca
- search.open.canada.ca/data

Common File Formats

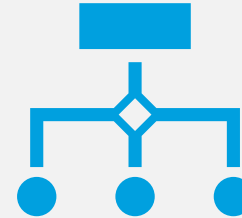
- CSV
- Spreadsheet
- JSON

What's in the metadata?



Every dataset should include:

- Title & description
- Source / publisher
- Last updated date
- Fields / variables
- License



Metadata should tell us:

- What is this data?
- When is it from?
- Who created it?
- What does each field mean?

LET'S PRACTICE!

Make sure your dataset is

- Recent enough
- Granular enough
- Relevant

Be skeptical now, or pay the price later.

Your challenge

Given a topic, find 3 datasets.

- Record the **source**, update **date**, and key **variables**
- **Why** is this dataset useful?

Limitations:

- Use at least one government portal.
- Datasets must be different.
- Be ready to justify! What question(s) could we answer?

Challenge Debrief



CHALLENGES



PATTERNS



SURPRISES

Dataset Evaluation Checklist

- ☐ **Purpose & Relevance:** What problem could this dataset help answer?
- ☐ **Source Credibility:** Who collected or published this data?
- ☐ **Timeliness:** How current is the data?
- ☐ **Granularity:** Is the level of detail appropriate?
- ☐ **Field Clarity:** Do you understand what each variable actually means?
- ☐ **Completeness:** Are important values missing?
- ☐ **Consistency:** Is the data internally consistent?
- ☐ **Bias & Limitation:** What does this dataset not capture?
- ☐ **Accessibility & Format:** Is the data usable with available tools?
- ☐ **Legal & Ethical Use:** Are there restrictions on use or sharing?

1-minute reflection

- Write down one insight you gained from exploring a portal.

I STARTED THE DAY WITH
LOTS OF PROBLEMS.

BUT NOW, AFTER HOURS
AND HOURS OF WORK,

I HAVE LOTS OF PROBLEMS
IN A *SPREADSHEET*.



What's next?

- Quiz #1